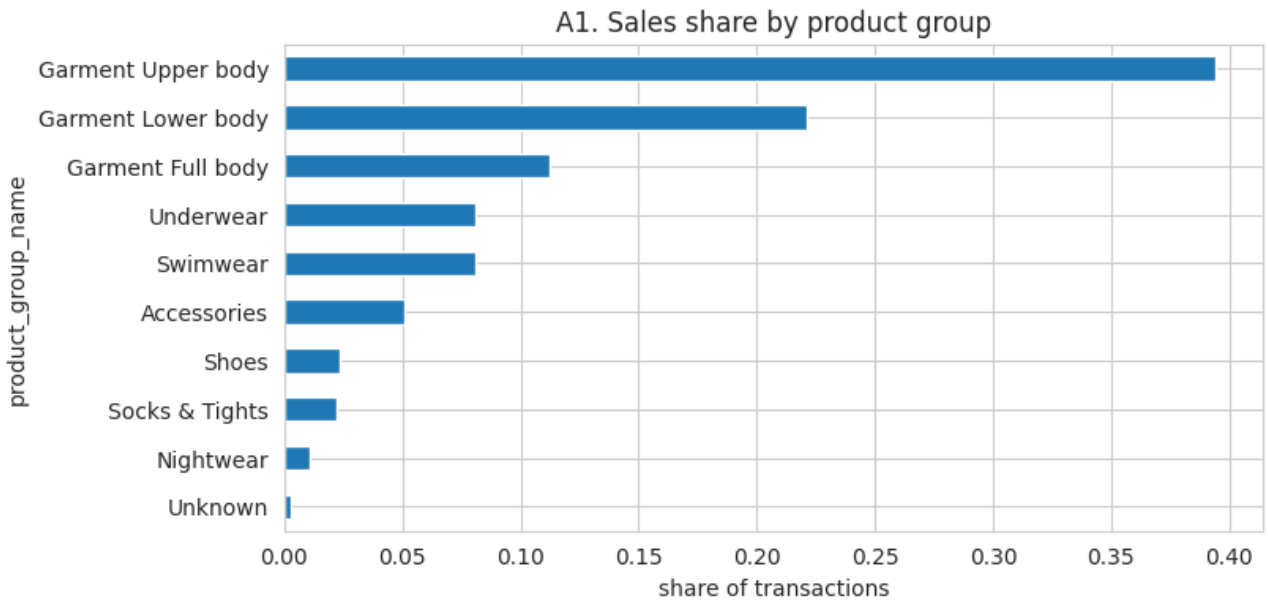
A1. Sales share by product group

Notebook cell 63



What it shows. The share of all transactions that belongs to each product group (top 10 groups).

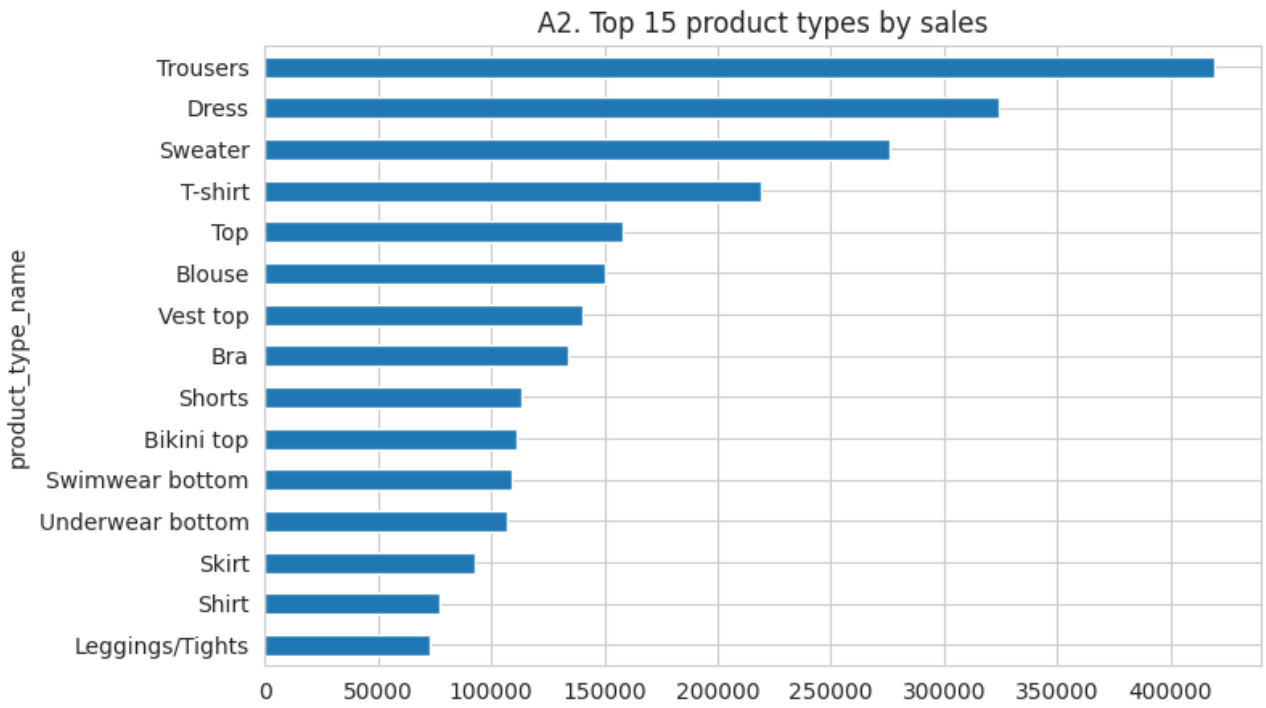
What we see in it

- Upper-body garments are the biggest group at about 39% of sales, then lower-body at about 22% and full-body at about 11%. These three groups together are about 72%.
- Underwear and Swimwear are about 8% each and Accessories about 5%.
- Shoes, Socks & Tights, Nightwear and Unknown are each about 2% or less.

Benefit. Shows that sales are concentrated in a few clothing groups, so popularity and group-level features will be driven by garments. Small groups have very few examples and need grouping or a fallback.

A2. Top 15 product types by sales

Notebook cell 65



What it shows. The 15 specific product types with the most transactions (counts, not shares).

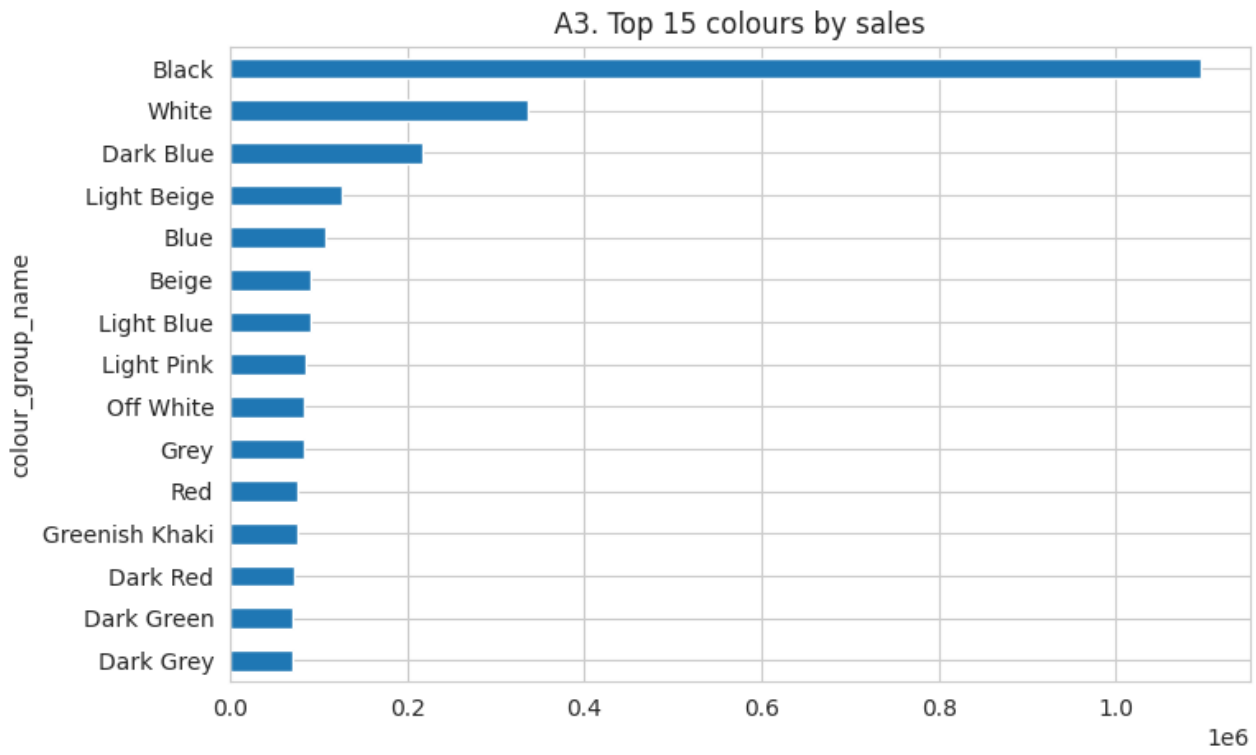
What we see in it

- Trousers is first (about 420,000 transactions), then Dress (about 325,000), Sweater (about 275,000) and T-shirt (about 220,000). These four are roughly 39% of all 3.16 million transactions (estimated from the bars).
- Swimwear (Bikini top, Swimwear bottom) and underwear types (Bra, Underwear bottom) also reach the top 15, with about 105,000 to 135,000 each.
- The bars decrease gradually; no single type dominates.

Benefit. Product type is a clear and useful item feature, and the top types make a ready-made 'popular items' fallback. The swimwear types in the top 15 hint at seasonality, which graphs C1 and C3 check.

A3. Top 15 colours by sales

Notebook cell 67



What it shows. The 15 colours with the most transactions.

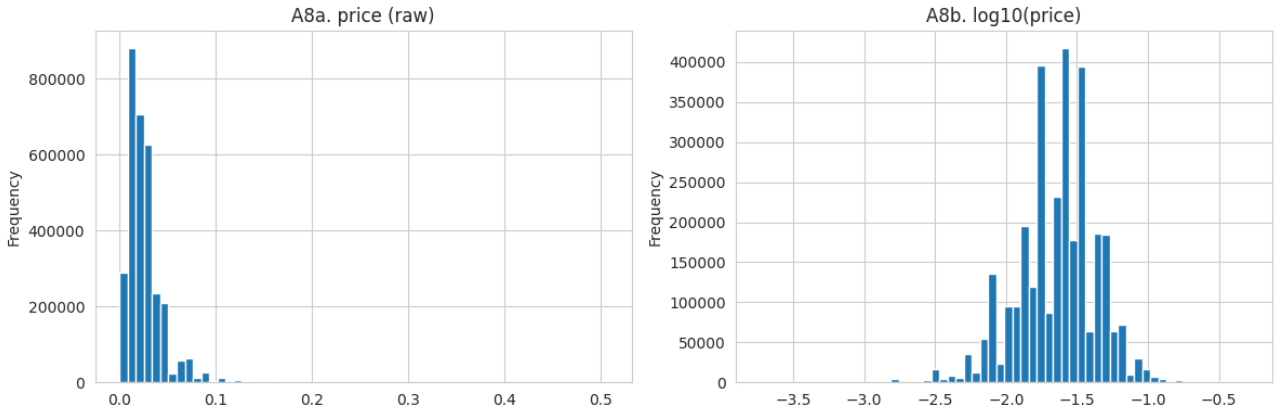
What we see in it

- Black is far ahead with about 1.1 million transactions (roughly 35% of all sales). White is next at about 335,000 and Dark Blue at about 215,000.
- After the top three, every colour is below about 125,000. Neutral and dark colours fill most of the list; Red only appears at 11th place.
- Comparison with the earlier catalogue count: Black is 21.5% of catalogue articles but about 35% of sales, so black items sell more than their catalogue share.

Benefit. Colour is very imbalanced, so on its own it separates items poorly. The long tail of rare colours should be grouped, and colour is best used together with other attributes.

A8. Price distribution (raw and log10)

Notebook cell 69



What it shows. Histograms of transaction price for all 3.16 million transactions: raw values on the left, log10 of price on the right.

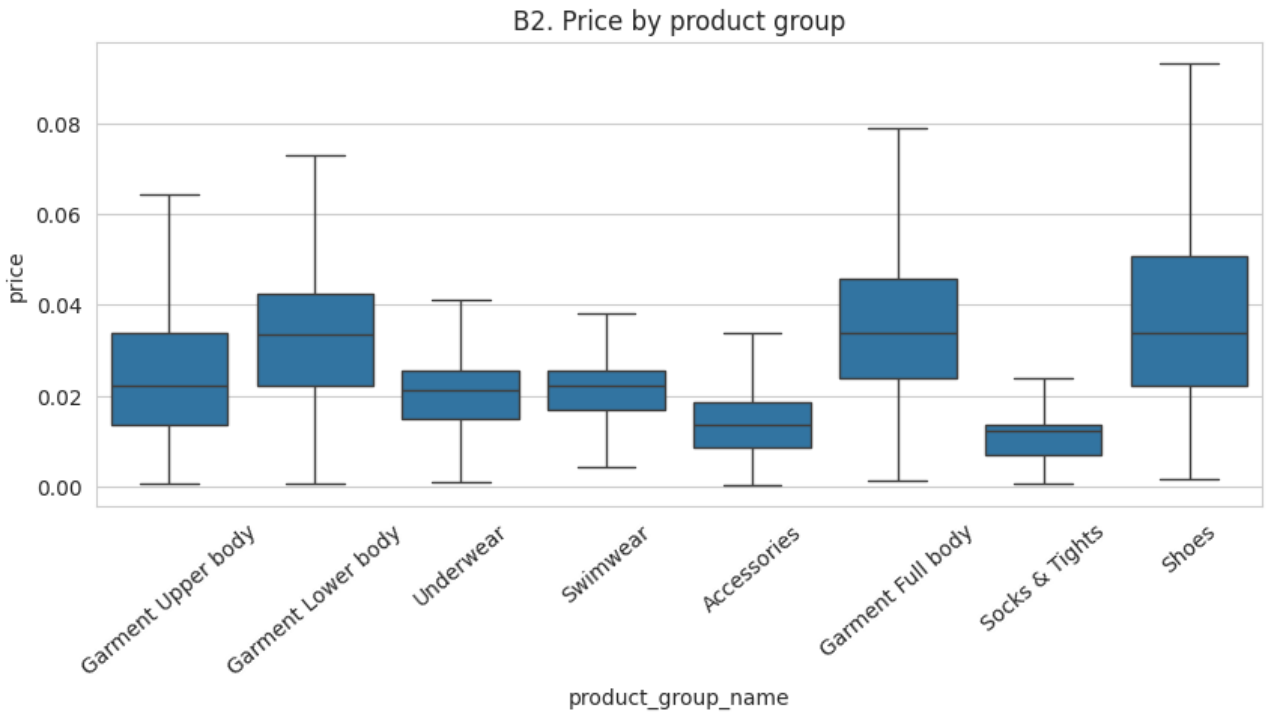
What we see in it

- Raw price is strongly right-skewed: most prices are below about 0.05, the tallest bar is near 0.015 to 0.02 (about 880,000 transactions), few are above 0.1, and the maximum is about 0.5.
- After log10 the shape is roughly bell-like, centred near -1.6 (a price of about 0.025), spread from about -3.7 to -0.3.
- The log plot has sharp spikes. This means prices cluster at fixed price points instead of varying smoothly.
- Prices are scaled values, not real currency, so small numbers are normal.

Benefit. Confirms price needs a log scale (or price bands) before modelling. The clusters suggest a 'price tier' feature could work well.

B2. Price by product group

Notebook cell 71



What it shows. Box plots of price for the 8 biggest product groups, drawn from a random 300,000-transaction sample. Outliers are hidden, so the whiskers are not the maximum.

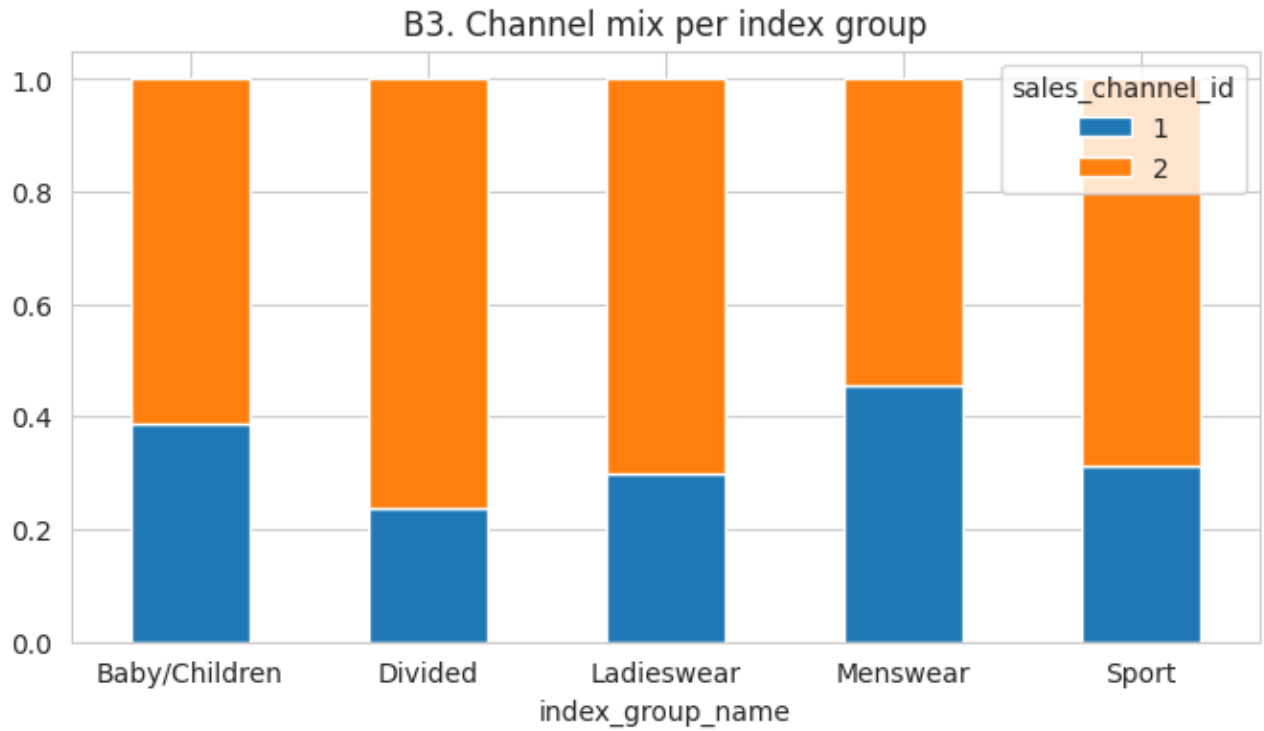
What we see in it

- Lower body, Full body and Shoes have the highest median price (about 0.034).
- Upper body, Underwear and Swimwear sit around 0.021 to 0.022.
- Accessories (about 0.014) and Socks & Tights (about 0.012) are the cheapest.
- Shoes have the widest spread (up to about 0.09); Socks & Tights have the narrowest.

Benefit. Price level depends on the product group, so a price should be judged inside its group (0.03 is expensive for an accessory but normal for shoes). This supports a 'price relative to group' feature.

B3. Channel mix per index group

Notebook cell 73



What it shows. For each segment (index group), the share of its transactions in sales channel 1 (blue) and channel 2 (orange). The notebook describes the channels as store vs online.

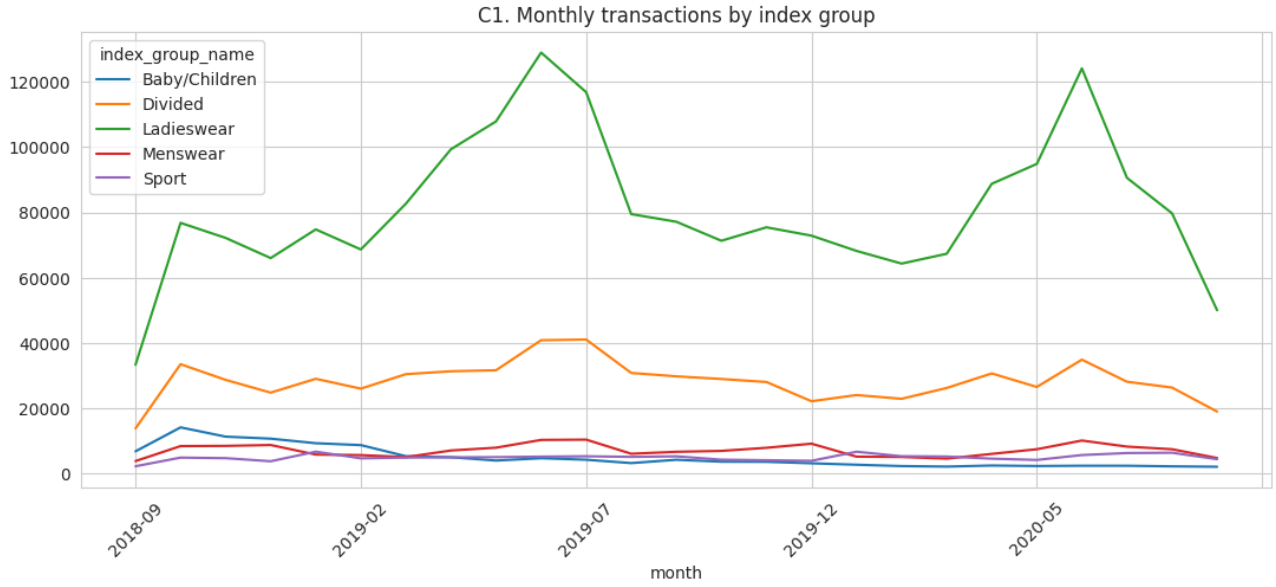
What we see in it

- Channel 2 is the larger channel in every segment.
- Channel 1 share: Menswear highest (about 46%), then Baby/Children (about 39%), Sport (about 31%), Ladieswear (about 30%) and Divided lowest (about 24%).
- So Divided is the most channel-2 segment, at about 76%. (The legend covers part of the Sport bar.)

Benefit. Segments have different channel habits, so channel is a useful feature and recommendations may need to be channel-aware.

C1. Monthly transactions by index group

Notebook cell 75



What it shows. Monthly transaction counts for each of the 5 segments, to compare trend and seasonality.

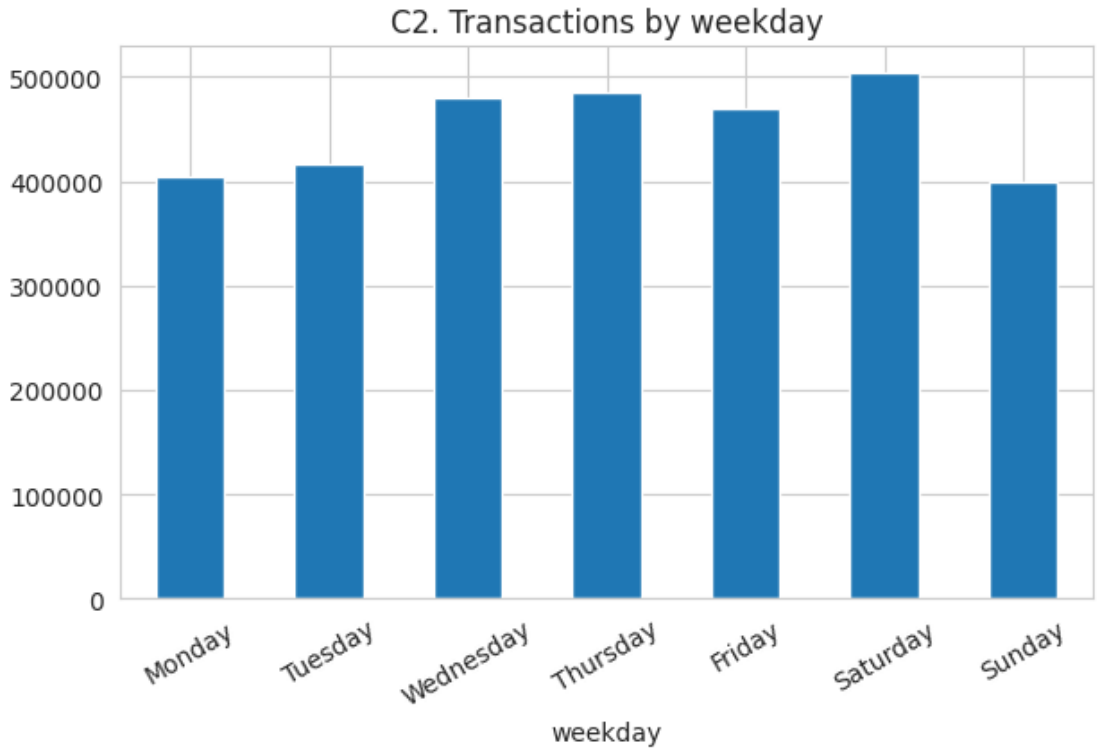
What we see in it

- Ladieswear is by far the largest and peaks around Jun 2019 (about 129,000) and Jun 2020 (about 124,000). Divided is second and peaks at about 41,000 in Jun to Jul 2019.
- Menswear stays around 5,000 to 10,000 with a small summer rise. Sport and Baby/Children stay low (about 2,000 to 7,000) with no clear seasonal peak in this chart.
- Ladieswear in Mar 2020 (about 64,000) is lower than Mar 2019 (about 83,000).
- The first (Sep 2018) and last (Sep 2020) months are partial, so their low values are not real drops.

Benefit. Seasonality is strong for the big segments and weak for the small ones, so seasonal features should be built per segment. Small segments give little signal.

C2. Transactions by weekday

Notebook cell 77



What it shows. Total transactions for each day of the week.

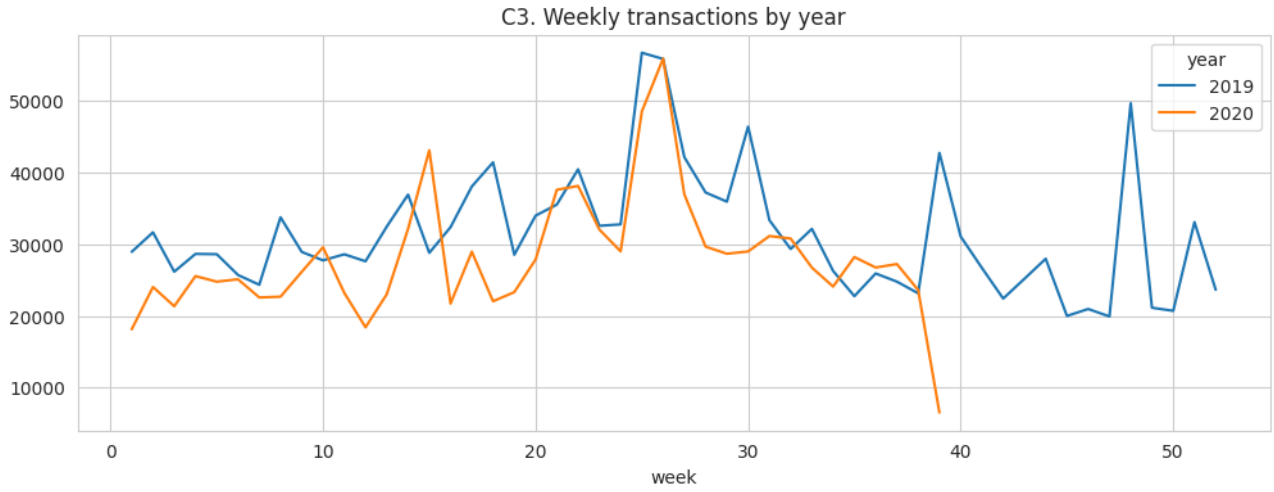
What we see in it

- Saturday is the busiest day (about 505,000), followed closely by Thursday and Wednesday (about 480,000) and Friday (about 470,000).
- Monday and Tuesday are about 405,000 to 415,000, and Sunday is the quietest (about 400,000).
- The busiest day is roughly 25% higher than the quietest.

Benefit. The weekday effect is modest but real, so a day-of-week feature is worthwhile. It also means validation windows should cover whole weeks so the weekday mix is balanced.

C3. Weekly transactions by year

Notebook cell 79



What it shows. Weekly transaction counts by week of the year, 2019 (blue) against 2020 (orange).

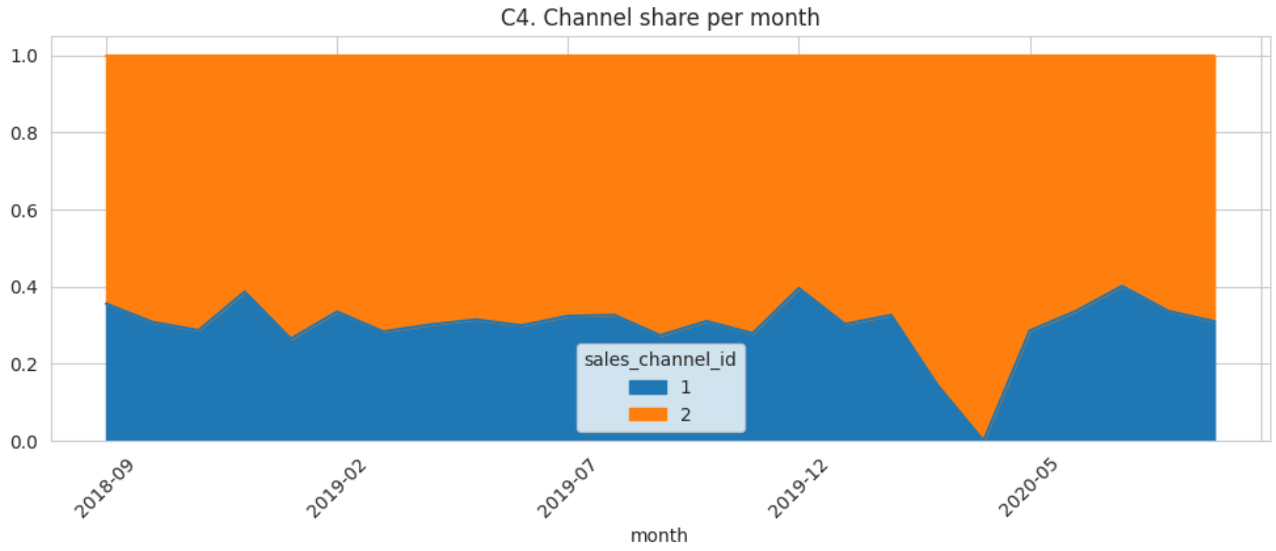
What we see in it

- Both years follow a similar shape: a rise to a peak around weeks 25 to 26 (about 57,000 in 2019 and about 56,000 in 2020), then a decline.
- 2020 is mostly below 2019, especially in weeks 1 to 14. 2020 has a spike at week 15 (about 43,000); 2019 has spikes at week 39 (about 43,000) and week 48 (about 50,000).
- The 2020 line ends at week 39 because the data stops on 22 Sep 2020. Its last point (about 6,600) is a partial week, not a real drop.

Benefit. Shows the same seasonal pattern repeating across years, which supports week-of-year features. The lower 2020 level warns that 2019 volumes cannot simply be copied to 2020.

C4. Channel share per month

Notebook cell 81



What it shows. The monthly share of transactions in channel 1 (blue) and channel 2 (orange), stacked to 100%.

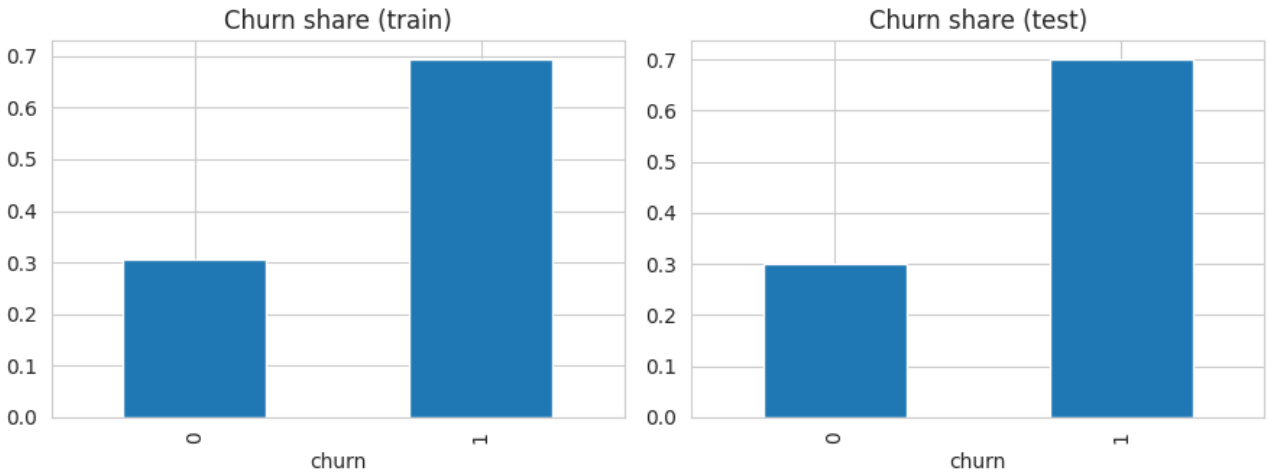
What we see in it

- In most months channel 1 is about 27% to 40% and channel 2 is about 60% to 73%.
- There is a sharp exception in spring 2020: channel 1 falls to about 15% in Mar 2020 and to almost 0% in Apr 2020, then returns to about 29% in May and about 40% by Jul 2020.
- The data does not say why. It may match store closures in that period, but that is an assumption.

Benefit. Detects a sudden change in behaviour. Spring 2020 differs from the rest of the data, so models trained on earlier months may not transfer, and any time-based validation should be aware of this period.

Churn 1. Target distribution (train and test)

Notebook cell 86



What it shows. The share of customers who churned (1) and did not churn (0) in the training snapshot (left) and the test snapshot (right). Churn means no purchase in the 28 days after the snapshot date.

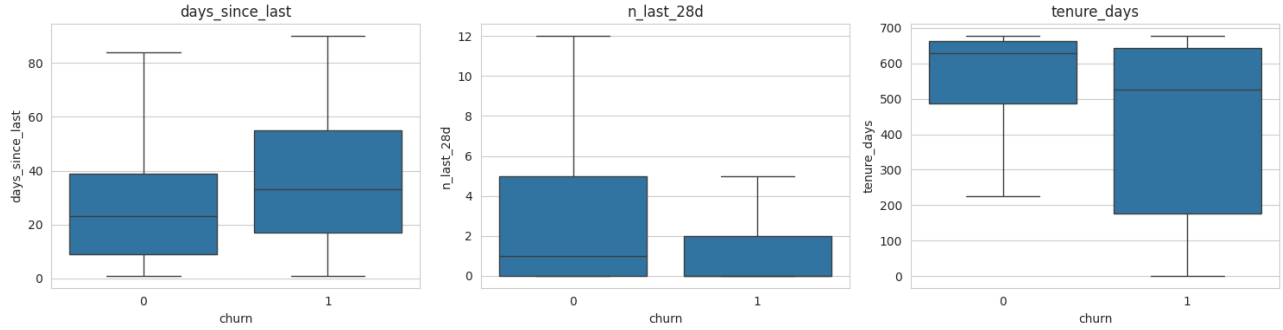
What we see in it

- Churn is the majority class: 69.5% of the 53,514 training customers and 70.1% of the 53,925 test customers (printed in the cell).
- The two snapshots have almost the same balance (about 30% stayed, about 70% churned).

Benefit. The target is imbalanced, so plain accuracy is misleading (predicting 'churn' for everyone already gives about 70%). Use PR-AUC, precision, recall and F1, and consider class weights. The similar rates in train and test show the time-based split is stable.

Churn 2. Key features by churn (boxplots)

Notebook cell 91



What it shows. Box plots of three features for non-churners (0) and churners (1), on a 50,000-customer sample. Outliers are hidden.

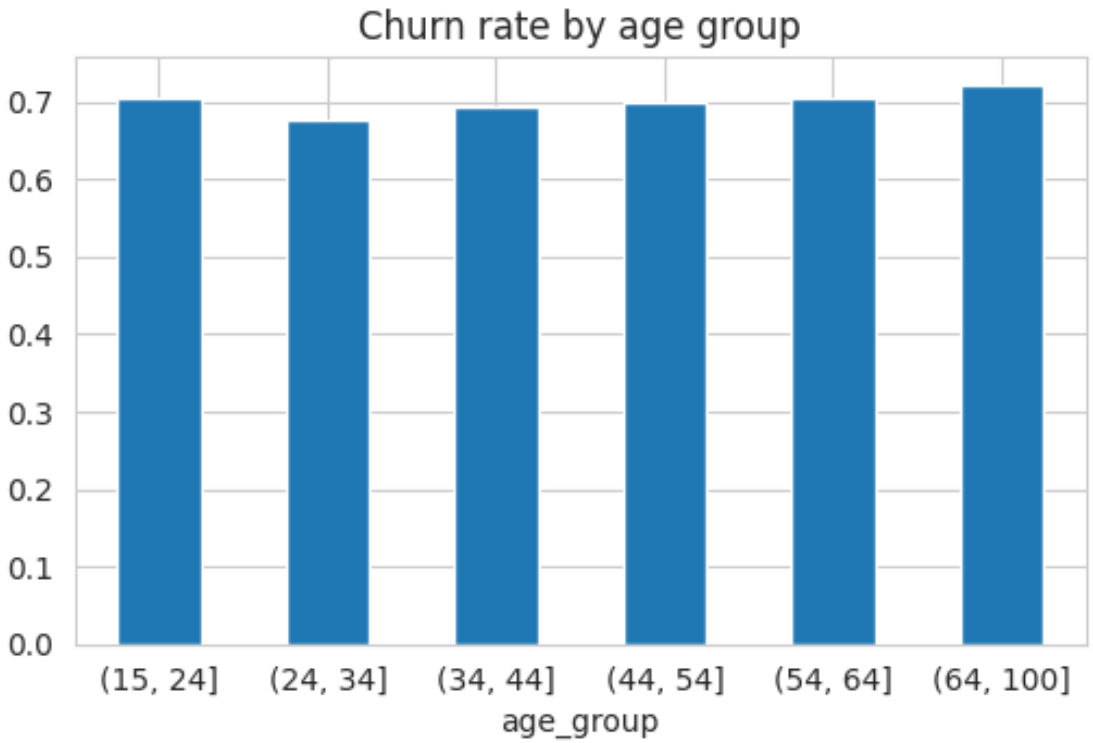
What we see in it

- days_since_last: churners have a higher median (33 days vs 23 days).
- n_last_28d (purchases in the last 28 days): the median is 0 for churners and 1 for non-churners; non-churners reach much higher values.
- tenure_days: churners have a lower median (527 vs 628) and a much wider spread, including many newer customers.
- The boxes overlap a lot, so no single feature separates the classes cleanly. The medians table in cell 90 agrees: n_purchases 42 vs 18, n_active_days 14 vs 6, total_spend 1.04 vs 0.45 (non-churn vs churn).

Benefit. Confirms that recency and recent activity carry signal, but only a moderate one, so a model needs several features together. These are the first features to keep.

Churn 3. Churn rate by age group

Notebook cell 92



What it shows. The average churn rate (0 to 1) for each age band.

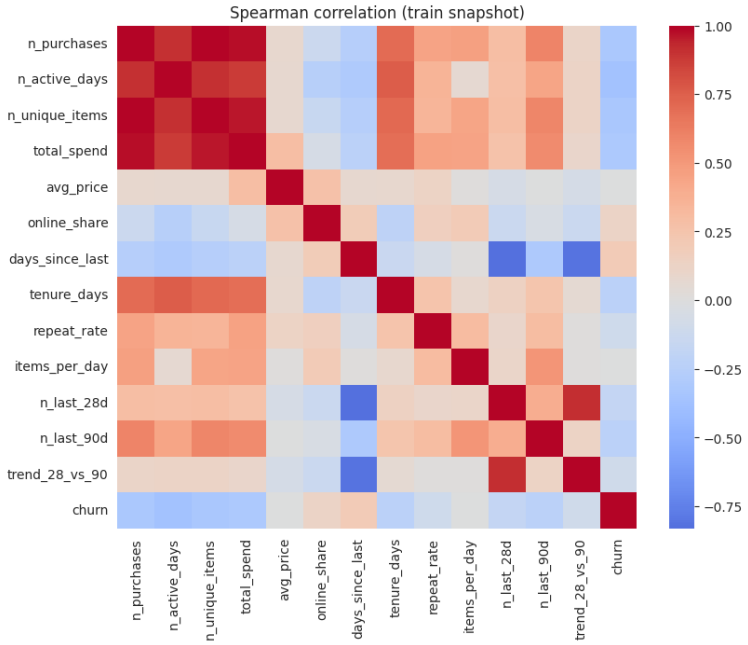
What we see in it

- All age groups are close, between about 0.67 and 0.72. The lowest is 25 to 34 (about 0.675) and the highest is 65 and over (about 0.72).
- The same cell prints churn by club status: ACTIVE 68.9% (52,072 customers), PRE-CREATE 91.3% (1,351), UNKNOWN 61.6% (86) and LEFT CLUB 60% (5).

Benefit. Age adds very little on its own, so it is a weak feature. PRE-CREATE members churn much more but are a small group; LEFT CLUB and Unknown are too small to draw any conclusion.

Churn 4. Spearman correlation heatmap

Notebook cell 94



What it shows. Rank correlation between all 13 numeric features and churn on the training snapshot (red = positive, blue = negative).

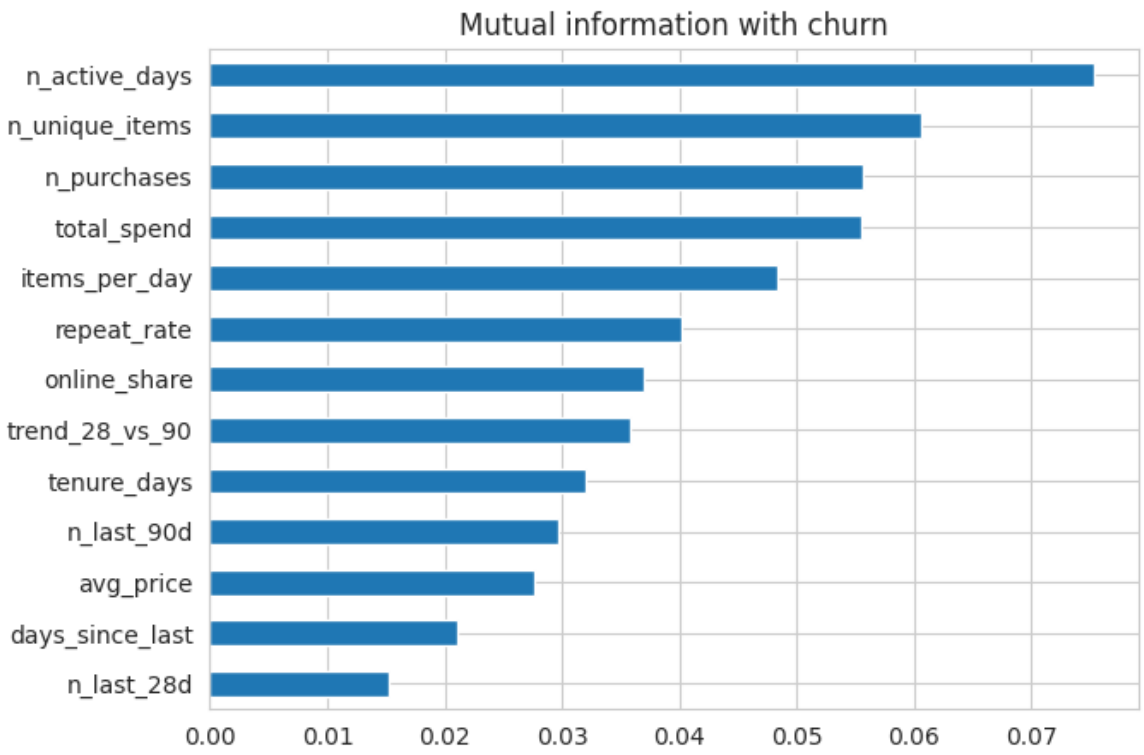
What we see in it

- n_purchases, n_active_days, n_unique_items and total_spend are almost perfectly correlated with each other (dark red block). n_last_28d and trend_28_vs_90 are also strongly linked.
- days_since_last is strongly negative with n_last_28d and trend_28_vs_90 (about -0.7 to -0.8).
- Against churn, the activity features are mildly negative (about -0.3), days_since_last is mildly positive (about +0.2); every correlation with churn is modest.
- VIF printed in cell 95: n_purchases 1,560, n_unique_items 1,076, n_active_days 255, items_per_day 67, repeat_rate 19.4, total_spend 10.2. All other features are below 10.

Benefit. Shows which features repeat the same information. Several of the high-VIF features should be dropped or combined (keep one of n_purchases, n_unique_items, n_active_days, total_spend) for linear models. Tree models cope with this better.

Churn 5. Mutual information with churn

Notebook cell 96



What it shows. How much each feature tells us about churn, including non-linear links (higher bar = more informative), on a 50,000-customer sample.

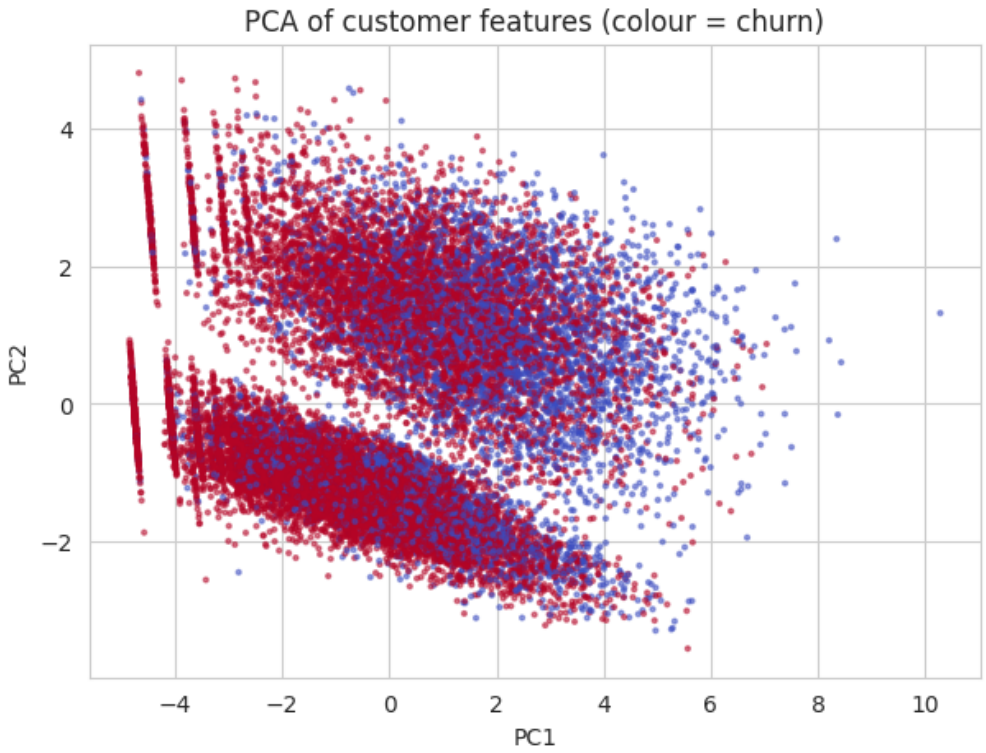
What we see in it

- The activity-volume features lead: n_active_days (about 0.075), n_unique_items (about 0.061), n_purchases and total_spend (about 0.055).
- Next come items_per_day, repeat_rate, online_share and trend_28_vs_90 (about 0.036 to 0.048).
- Recency features rank lowest: days_since_last (about 0.021) and n_last_28d (about 0.015), even though the boxplots show a difference in their medians.
- All values are small (below 0.08), so no feature is strongly predictive alone.

Benefit. Gives a ranking for feature selection that does not assume a straight line. It also shows the notebook's expectation (recency as the strongest signal) is not what this measure finds, so it is worth checking again with the models.

Churn 6. PCA of customer features (coloured by churn)

Notebook cell 98



What it shows. The 13 features reduced to 2 components (PC1 and PC2) for 20,000 customers. Red = churn, blue = no churn.

What we see in it

- Explained variance printed in the cell: PC1 39.6%, PC2 18.3%, PC3 12.3%, PC4 9.1%, PC5 6.2%. The first two components hold about 58% and the first five about 85%.
- Churners (red) are concentrated on the left (low PC1, the low-activity side). Non-churners (blue) spread to the right (high PC1).
- The middle is heavily mixed, so the classes only partly separate.
- The plot shows two diagonal bands and vertical streaks on the far left; the notebook does not explain them (the streaks look like customers with very small, discrete feature values).

Benefit. Shows that a few components capture most of the variation, which fits the redundancy found by VIF. It also shows churn is only partly separable by these features, so expect a moderate model score and judge it with PR-AUC.